DILLAN IMANS

Medical AI Research Engineer | Medical Imaging & Computer Vision
Relocating to London, UK (Oct 2026)
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SUMMARY

Research Engineer with first-author publications at IEEE BIBM 2025 and Diagnostics, specializing in medical image segmentation, multimodal learning, and clinical AI deployment across brain MRI, retinal imaging, and surgical vision. Incoming MSc AI for Biomedicine and Healthcare at UCL.

EDUCATION

MSc AI for Biomedicine and Healthcare, University College London — **Oct 2026 - Oct 2027 (Expected)**

B.S. Computer Science & Engineering, Sungkyunkwan University — **Aug 2022 - Jun 2026**
95.9 / 100 · Dean's List 2023 · Academic Excellence Scholarship

EXPERIENCE

AI Research Engineer · Superintelligence Lab SKKU, South Korea — **Oct 2024 - Present**

- Developed unsupervised domain adaptation framework for brain tumor segmentation across MRI domain shift, outperforming benchmarks by ~5–10% AUC (First Author, IEEE BIBM 2025)
- Built end-to-end retinal analysis system covering optic disc/cup segmentation, vessel analysis, and glaucoma detection; optimized for clinical deployment via ONNX with low-latency inference, integrated with industry-partnered screening platform.
- Designed cross-modality knowledge distillation framework for hypertension risk prediction from unpaired MRI, ECG, and fundus images (First Author, preprint available)

AI Research Engineer · Labren CUHK, Hong Kong — **Jun 2024 - Present**

- Designed preprocessing infrastructure and annotation QA pipeline for the first large-scale surgical action recognition dataset (11,915 clips)
- Collaborated with NVIDIA on multimodal surgical robotics dataset construction, supporting pipeline architecture and data standardization
- Co-authored 3 preprints in submission at ECCV 2026 and npj Digital Surgery on surgical action planning, skill assessment, and robot policy learning

AI Research Engineer · Infolab SKKU, South Korea — **Feb 2024 - Oct 2024**

- Developed explainable multi-layer dynamic ensemble framework for depression detection and severity prediction from behavioral features (First Author, Diagnostics 14(21) 2385, Oct 2024)
- Designed interpretable clinical ML pipeline with feature-level explainability for mental health screening applications

PUBLICATIONS

Full list: scholar.google.com/citations?user=o-TtUq0AAAAJ

- Imans D. et al. "Unsupervised Domain Adaptation with SAM-RefiSeR for Enhanced Brain Tumor Segmentation." IEEE BIBM 2025. (First Author)
- Imans D. et al. "Explainable Multi-Layer Dynamic Ensemble Framework Optimized for Depression Detection and Severity Assessment." Diagnostics, 14(21), 2385 (2024). (First Author)
- 1 first-author preprint on arXiv · 3 co-authored preprints under review

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript
ML / AI: PyTorch, MONAI, HuggingFace, scikit-learn, OpenCV, wandb, TensorBoard
Medical / Bio: NiBabel, SimpleITK, pydicom, AlphaFold, BLAST
Tools: Git, Linux, Docker, AWS
Spoken Languages: English & Indonesian (Fluent), Mandarin & Korean (Conversational)